SIMATS Online Programming Lab

DS PROGRAMMING

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QUESTIONS 10 / 10 completed

Array Rotation

ARRAY ROTATION

"Implement array rotation (left/right) for arrays of size 10 by K positions without using extra space. Use a reversal algorithm. The program must validate K (0 ≤ K < N), show original→rotated→reversal steps visually, compute the sum difference, and support multiple rotations.
Sample Input/Output:
Enter 10 numbers: 1 2 3 4 5 6 7 8 9 10
Rotate Left (L)/Right (R): L
By positions: 3
Original: [1 2 3 4 5 6 7 8 9 10]
Step1: Reverse 0-2 → [3 2 1 4 5 6 7 8 9 10]
Step2: Reverse 3-9 → [3 2 1 10 9 8 7 6 5 4]
Step3: Reverse all → [4 5 6 7 8 9 10 1 2 3]
Sum difference: 0"

PROGRAM EDITOR

C

Run Save

```
#include <stdio.h>
int main(){
  int a[10],k,i,j,temp;
  printf("enter 10 no:\n");
  for(i=0;i<10;i++){
    scanf("%d",&a[i]);
  }
  printf("enter k:");
  scanf("%d",&k);
  for(i=0;i<k;i++){
    temp=a[0];
    for(j=0;j<9;j++)
      a[j]=a[j+1];
    a[9]=temp;
  }
  printf("\nrotated array:\n");
  for(i=0;i<10;i++){
    printf(" %d",a[i]);
  }
  return 0;
}
```

OUTPUT

```
enter 10 no:
enter k:rotated array:
 4 5 6 7 8 9 10 1 2 3
```

INPUT

```
1 2 3 4 5 6 7 8 9 10
3
```